ANVESH SAI VEMURI
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LinkedIn| GitHub| Portfolio

PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience designing and optimizing scalable batch and streaming data pipelines, distributed data processing systems, data lakes, lakehouse architectures, and cloud data platforms using Python, SQL, PySpark, Apache Spark, AWS, Databricks, Snowflake, and BigQuery. Experienced in building production ETL/ELT pipelines, workflow orchestration, data modeling, data quality frameworks, API-driven ingestion, and high-volume data processing across enterprise environments. Hands-on experience with Amazon S3, AWS Glue, Lambda, Redshift, EMR, Athena, Lake Formation, Airflow, dbt, Kafka, Delta Lake, and Spark Streaming, with exposure to AI/ML-ready data pipelines, LLM data processing, vectorized data workflows, and RAG-oriented data preparation.

TECHNICAL SKILLS

- **Programming:** Python, SQL, Java, Pandas, NumPy
- **Data Engineering:** ETL/ELT, Data Pipelines, Apache Spark, PySpark, Databricks, Delta Lake, Apache Airflow, dbt, Data Integration, Data Transformation, REST APIs, API-based Ingestion
- **Cloud & Data Platforms:** AWS, Amazon S3, AWS Glue, AWS Lambda, Amazon Redshift, Amazon Athena, Amazon EMR, AWS Lake Formation, GCP, BigQuery, Snowflake
- **Data Warehousing & Modeling:** Data Warehousing, Dimensional Data Modeling, Star Schema, Data Lakes, Lakehouse Architecture, Hive, HDFS
- **Streaming & Messaging:** Kafka, Spark Streaming
- **GenAI & AI/ML Data Engineering:** LLM Data Processing, AI/ML Data Pipelines, Embedding Pipelines, Vectorized Data, RAG Data Preparation, Unstructured Data Processing, Amazon Bedrock
- **Data Quality & Governance:** Data Quality, Data Validation, Schema Validation, Data Reconciliation, Data Governance, Pipeline Monitoring
- **DevOps & Tools:** Git, GitHub, Jenkins, Docker, Azure DevOps, Linux, CI/CD, Alteryx, Jupyter Notebook
- **Analytics:** Tableau, Power BI

PROFESSIONAL EXPERIENCE

Citibank, USA— Data Engineer – GenAI & Data Platforms **Sep 2026 – Present**

- Engineered Python and SQL data pipelines to ingest, cleanse, normalize, and transform structured and unstructured enterprise data for downstream analytics and AI/ML workloads.
- Built scalable data-processing workflows for LLM applications, preparing enterprise documents and datasets for chunking, metadata enrichment, embedding generation, and retrieval-oriented use cases.
- Developed automated data pipelines integrating AWS S3, AWS Glue, Lambda, Databricks, and Python to process unstructured content and deliver validated datasets for GenAI applications.
- Implemented RAG-oriented data preparation workflows, including document preprocessing, data validation, metadata management, and vectorization pipelines to improve the quality and consistency of enterprise knowledge data.
- Applied data-quality and validation frameworks to AI/ML datasets, implementing schema checks, completeness validation, duplicate detection, and reconciliation controls before downstream model consumption.
- Optimized Spark and PySpark transformations for large-scale structured and unstructured datasets, improving pipeline scalability and reducing processing overhead across AI-ready data workloads.
- Orchestrated production data workflows using Apache Airflow and automated monitoring, logging, and failure-handling processes to improve pipeline reliability and operational visibility.

Publicis Groupe (Zenith Media Services) — Data Engineer **Sep 2024 – Aug 2026**

- Engineered Python and SQL ETL pipelines ingesting and transforming advertising and eCommerce data from 20+ external platforms across BigQuery, Redshift, and Databricks, reducing reporting latency by 40% and supporting analytics for 6+ enterprise clients.
- Migrated 50+ production Alteryx workflows to Databricks, re-engineering end-to-end lakehouse pipelines, transformations, and orchestration, reducing execution time by 35%.
- Optimized complex SQL transformations, stored procedures, and incremental data-processing workflows across cloud data warehouses, improving query performance by 30% while supporting high-volume analytics workloads.
- Implemented automated data-quality checks, reconciliation logic, schema validation, and pipeline monitoring across production ETL workflows, reducing manual QA effort by 10+ hours per week and improving reliability of downstream reporting.
- Integrated data from 20+ platforms, including Pinterest, DoubleVerify, DV360/YouTube, Amazon, Walmart, Meta, TikTok, Snapchat, Reddit, and TTD, standardizing campaign and performance data used across 100+ downstream dashboards.

JPMorgan Chase, USA — Data Engineer **Sep 2023 – Aug 2024**

- Developed distributed PySpark ETL pipelines processing 20M+ financial and transactional records daily, improving processing performance by 45% while supporting downstream analytics and reporting.
- Built 100+ SQL and dbt transformations in Snowflake and Hive for large-scale financial and transactional datasets, implementing reusable Star schemas and improving analytical query performance by 35%.
- Optimized distributed Spark and Hive workloads using partition-aware processing, Parquet storage, and Spark SQL tuning to improve performance and scalability across large enterprise datasets.
- Automated and maintained 25+ Apache Airflow workflows for enterprise data pipelines, improving orchestration, monitoring, and data reliability while reducing manual intervention by 12 hours per week.
- Implemented automated dbt tests and data quality checks to validate schema integrity, transformation logic, and downstream reporting accuracy across financial datasets.

Cognizant, India — Data Engineer **Aug 2020 – Jul 2021**

- Developed 15+ custom Apache Airflow operators in Python integrating Snowflake, Slack, and Tableau workflows, automating enterprise data pipelines and improving monitoring and operational efficiency.
- Designed and implemented Amazon Redshift data warehouse schemas and ETL pipelines using Python and SQL, improving query performance by 35% and reducing report generation time by 40% for high-volume analytics workloads.
- Built event-driven Spark Streaming and AWS Lambda pipelines processing 5M+ real-time events daily, enabling low-latency anomaly detection and improving operational responsiveness.
- Developed scalable ETL pipelines using AWS Glue, Amazon S3, AWS Lambda, Boto3, Databricks, and PySpark, leveraging Boto3 for automated S3 file ingestion and object management, reducing cloud compute costs by 25% while improving pipeline efficiency and scalability.

EDUCATION

PhD, Technology and Artificial Intelligence **Apr 2029**
Southwest Baptist University
Master of Science, Computer Information Systems **Dec 2023**
New England College

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **AWS Certified Data Engineer – Associate — In Progress**
- **HackerRank SQL (Advanced) Certificate — HackerRank** **Jan 2026**
- **Databricks Academy – Advanced Data Engineering Training**
  Unity Catalog, Lakeflow Jobs, Lakeflow Connect, Spark Declarative Pipelines **Jan 2025**